Lecture 15 (B14): Krusell–Smith and Young's Method

Heterogeneous agents, the KS insight, non-stochastic distribution updates

Simon Scheidegger

HEC, University of Lausanne

Outline

1. **Motivation: Why Track Distributions?**
   From representative agents to heterogeneous agents; the infinite-dimensional state space.
2. **Young's (2010) Non-Stochastic Simulation**
   Histogram representation, mass redistribution, mean preservation, full algorithm.
3. **Benchmark: Krusell–Smith (1998)**
   Approximate aggregation, the forecasting-rule benchmark, and where Young fits.
4. **DEQN Teaching Implementation**
   Appendix A.5 endowment economy, histogram encoding, network architecture, training loop.
5. **Alternative Deep-Learning Approaches to KS**
   All-in-one DL (Maliar–Maliar–Winant, 2021) and DeepHAM (Han–Yang–E, 2024).

Part I

Motivation: Why Track Distributions?

From Representative Agent to Heterogeneous Agents

Days 2–4 recap:

- Brock–Mirman: single agent, single capital stock
- IRBC: N countries, but a social planner with complete markets
- State: $(k^1, \ldots, k^N, z^1, \ldots, z^N)$—finite-dimensional

$\Rightarrow$ All agents are "representative."

Reality:

- Agents differ in wealth, income, employment status
- Incomplete markets: cannot insure against idiosyncratic risk
- Aggregate shocks affect agents differently

Key questions:

- How does aggregate uncertainty affect inequality?
- How does the wealth distribution feed back into prices?

Why Distributions Matter

Policy implications:

- Fiscal stimulus: who spends the transfer?
- Monetary policy: affects borrowers and savers differently
- Wealth distribution shapes aggregate demand
- Consumption response depends on position in distribution

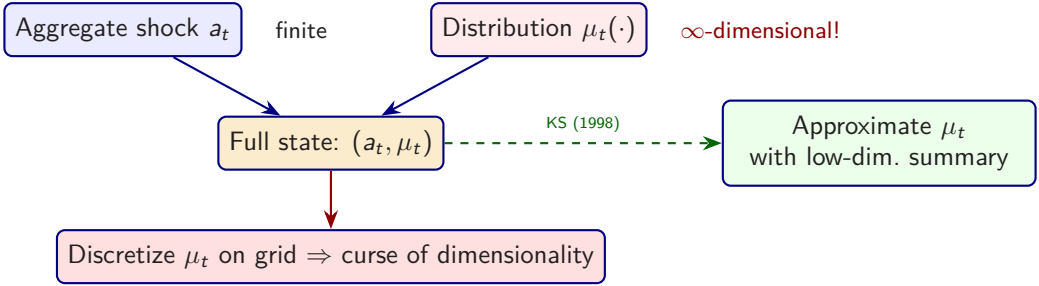
Agents near the borrowing constraint have high MPC; wealthy agents save most of a windfall.

Mathematical challenge:

- The entire distribution μ_t is a state variable
- μ_t is a measure over individual states
- Infinite-dimensional object!
- Cannot put it on a grid without the curse of dimensionality

$\Rightarrow$ Need clever methods to approximate μ_t.

The Infinite-Dimensional State Space Problem



- Traditional methods: discretize the distribution $\Rightarrow$ exponential cost
- Key insight (Krusell–Smith): low-order moments may suffice

Two Approaches to Heterogeneous Agents

Approach 1: Finitely Many Agents

Track N agents explicitly: $(k_1, \ldots, k_N, \varepsilon_1, \ldots, \varepsilon_N)$

Pro:

- Exact—no approximation of μ_t

Con:

- Permutation problem: agents are exchangeable
- State space grows with N
- Scales poorly for large N

Maliar, Maliar & Valli (2010)

Approach 2: Continuum of Agents

Track the distribution μ_t directly

Pro:

- Law of large numbers $\Rightarrow$ no sampling noise
- No permutation issue
- Distribution is order-invariant

Con:

- Must approximate μ_t (infinite-dimensional)

Krusell & Smith (1998), Young (2010)

$\Rightarrow$ **This lecture:** Approach 2 with Young's histogram method + DEQNs.

The Krusell–Smith Insight (1998)

Key insight: the mean of the wealth distribution is *nearly sufficient* to predict future prices.

Approximate bounded rationality: agents forecast using $m_1 = \int k \, d\mu$ (mean capital).

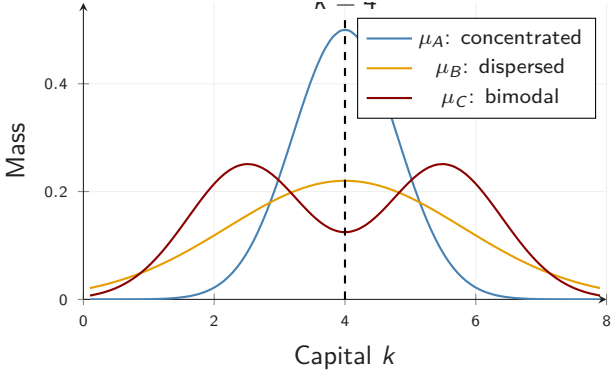
Log-linear forecasting rule:

$$\boxed{\hat{H}(m_1, a) = \exp\big(A(a) + B(a) \log m_1\big)}$$

- Coefficients $A(a), B(a)$ depend on the aggregate shock a
- Inada conditions ensure capital remains positive and logs are well-defined
- $R^2 > 0.999$ in practice—remarkably accurate!

But how do we simulate the distribution forward to compute realized m_1?

Why the Mean Is (Nearly) Sufficient



Bond prices:

$p(\mu_A) = 0.981$
$p(\mu_B) = 0.979$
$p(\mu_C) = 0.980$

Nearly identical!

Same mean $\bar{k} = 4$

$\Rightarrow$ similar prices

Three very different distributions that share the same mean capital $\bar{k} = 4$. In the KS economy, the resulting bond prices differ by $< 0.3\%$. This is why the mean is a nearly sufficient statistic for forecasting—and why approximate aggregation works so well.

But How Do We Simulate the Distribution?

Given policy functions, need to compute the next-period distribution μ_{t+1}.

Option A: Monte Carlo simulation

- Draw N agents, simulate forward, compute sample moments
- Problem: sampling noise $\sim O(1/\sqrt{N})$
- Need very large N ($> 50{,}000$) for accurate means
- Noise accumulates over time

Option B: Non-stochastic simulation (Young, 2010)

- Track the histogram directly on a fixed grid
- No sampling error—deterministic update
- Preserves the mean exactly
- Computationally fast

$\Rightarrow$ This lecture focuses on **Option B**: Young's (2010) method

Where Are We?

1. Motivation: Why Track Distributions? ✓
2. ⇒ Young's (2010) Non-Stochastic Simulation
3. Benchmark: Krusell–Smith (1998)
4. DEQN teaching implementation
5. Alternative deep-learning approaches to KS

Part II

Young's (2010) Non-Stochastic Simulation

The Core Idea

Instead of simulating a panel of agents $\longrightarrow$ simulate the distribution directly.

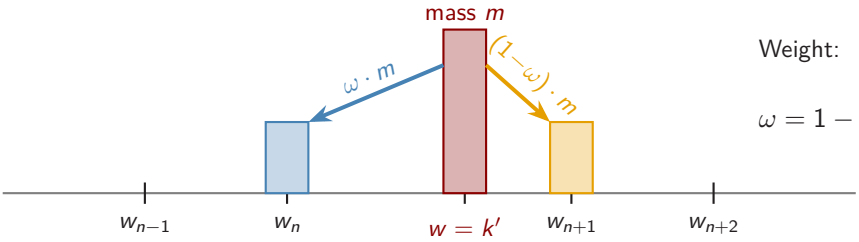
Setup:

1. Choose a fixed grid: $[w_1, w_2, \ldots, w_N]$
2. Represent the distribution as a histogram: mass $G(w_i)$ at each grid point
3. Update the histogram deterministically using:
   - Policy functions $k' = g(k, \varepsilon, m_1, a)$
   - Transition probabilities $\pi_{\varepsilon'|\varepsilon,a}$

Key insight

When k' falls between grid points, redistribute the mass to the two nearest grid points using linear interpolation weights. This preserves the mean exactly.

Mass Redistribution: The Key Operation



Weight:

$$\omega = 1 - \frac{w - w_n}{w_{n+1} - w_n}$$

- Mass m at $w \in [w_n, w_{n+1}]$ is split between the two bracketing grid points
- Weight ω: fraction to w_n (closer $\Rightarrow$ more mass)
- Weight $1 - \omega$: fraction to w_{n+1}

Lottery View: Why "Non-Stochastic"

The redistribution *is* a lottery

Every agent at (k, ε) whose policy choice k' lands between k_n and k_{n+1} is reassigned by a biased coin:

$$k_{t+1} = \begin{cases} k_n & \text{with probability } \omega, \\ k_{n+1} & \text{with probability } 1-\omega, \end{cases} \qquad \omega = 1 - \frac{k' - k_n}{k_{n+1} - k_n}.$$

Monte Carlo panel

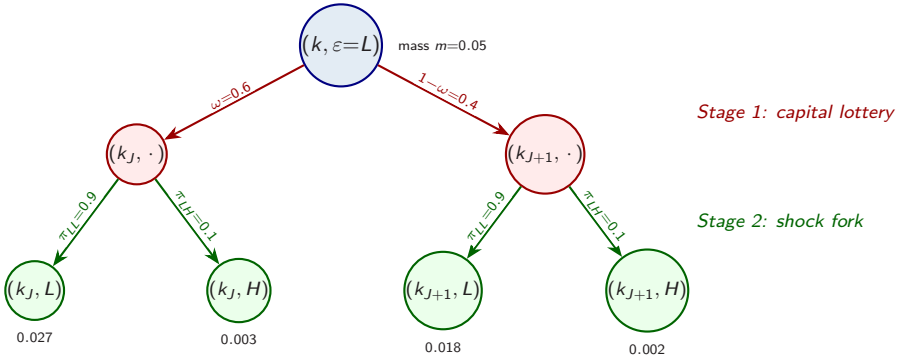
Each of N agents draws the lottery once.
Empirical histogram converges at rate $\mathcal{O}(N^{-1/2})$.
Sampling noise contaminates aggregates.

Young's method

Integrates the lottery analytically.
Equivalent to running $N = \infty$ agents.
Zero noise; mean preserved exactly.

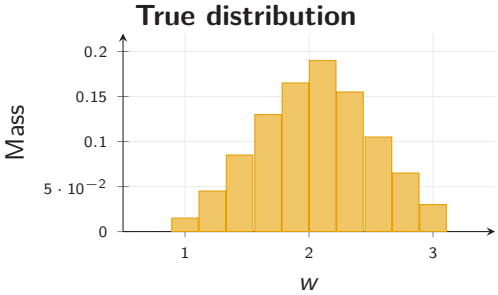
The same trick is applied to the shock transitions $\pi_{\varepsilon\varepsilon'}$: each piece of mass goes into *all* reachable next-period ε' in proportion to π, instead of one realisation being drawn.

One Cell Splits Into Four (Young, Fig. 1)

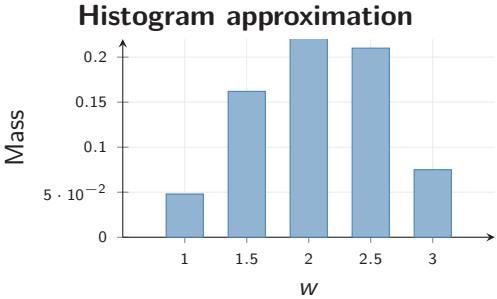


Verification: $0.027 + 0.003 + 0.018 + 0.002 = 0.05 = m$ ✓. Repeat for every active source bin and accumulate the leaves to obtain G_{t+1}.

Example: Multi-Point Distribution



10 points with truncated normal masses



5-point grid approximation

	True	Approximation
Mean	2.0	2.0 (exact!)
Variance	0.347	0.384 (approx.)

Key Property: Mean Preservation

The weight ω is forced by mean preservation

ω is the unique value for which the two-point split $\{\omega \cdot w_n, (1-\omega) \cdot w_{n+1}\}$ has expected value equal to the off-grid policy w. Solving $\omega \cdot w_n + (1-\omega) \cdot w_{n+1} = w$ for ω gives back $\omega = 1 - (w - w_n)/(w_{n+1} - w_n)$ in one line:

$$\omega \cdot w_n + (1-\omega) \cdot w_{n+1} \;=\; w_n + (1-\omega)(w_{n+1} - w_n) \;=\; w_n + (w - w_n) \;=\; w. \quad \checkmark$$

- Total mass preserved automatically: $\omega + (1-\omega) = 1$.
- By linearity of expectation, the unconditional mean of G_{t+1} equals $\mathbb{E}[k']$ *exactly*, provided no mass leaks past the grid endpoints.

Caution

Higher moments (variance, percentiles) are only *approximated*—finer grid = better approximation. Linear interpolation is exact for linear functions but smooths nonlinear ones.

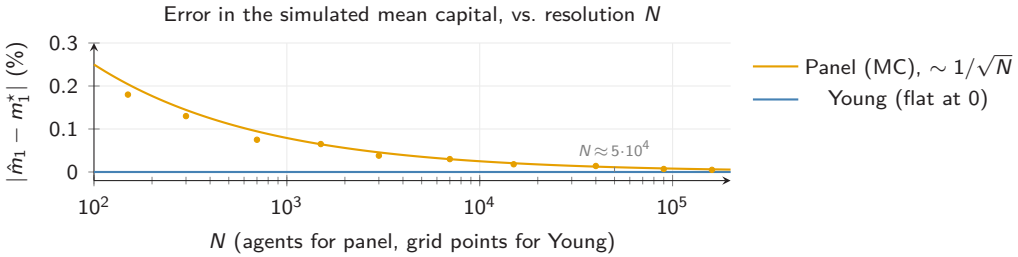
Why Not Just Use a Panel?

	Young's Method	**Panel Simulation**
Sampling noise	None	$O(1/\sqrt{N})$
Mean preservation	Exact	Approximate
Agents/points needed	$\sim$100–5000 grid pts	$>$50,000 agents
Speed	Fast (deterministic)	Slower (stochastic)
Memory	Fixed grid	Growing with N
Reproducibility	Deterministic	Seed-dependent

- Mean preservation is the headline property: each Young update reassigns mass deterministically using the lottery weight ω, so the new mean equals the policy-implied mean exactly (boundary clipping aside).
- In a Monte-Carlo panel the same lottery is *drawn once per agent*, so the realized mean is noisy at rate $1/\sqrt{N}$.

$\Rightarrow$ The next slide quantifies the gap.

Convergence: Young vs. Monte-Carlo Panel



What is plotted. Deviation of the simulated mean capital $\hat{m}_1$ from its true value $m_1^\star$, as the resolution N grows. Orange dots: realized Monte-Carlo errors of a panel of N agents, scattered around the $1/\sqrt{N}$ envelope. Blue line: Young's deterministic update, which preserves the mean exactly at *every* grid resolution.

Implication. To match Young's accuracy at $\sim$1,000 grid points, a Monte-Carlo panel needs $\gtrsim$ 50,000 agents (an order of magnitude more state), and the residual noise still contaminates the OLS step that updates the KS forecasting rule.

The Algorithm: Setup

Ingredients:

- **Asset grid:** $[k_1, k_2, \ldots, k_{N_k}]$ for individual capital/asset holdings
- **Idiosyncratic states:** $\varepsilon \in \{e, u\}$ (employed/unemployed)
- **Histogram:** $G_t(k_i, \varepsilon_j)$ = mass of agents at grid point k_i with shock ε_j
- **Total bins:** $N_k \times N_\varepsilon$ histogram entries

Normalization:

$$\sum_{i=1}^{N_k} \sum_{j=1}^{N_\varepsilon} G_t(k_i, \varepsilon_j) = 1$$

First moment (mean capital):

$$m_1 = \sum_{i=1}^{N_k} \sum_{j=1}^{N_\varepsilon} G_t(k_i, \varepsilon_j) \cdot k_i$$

The Algorithm: One Forward Step

Algorithm: Histogram Update at Time t

Input: Current histogram G_t, policy $g(\cdot)$, transition $\pi_{\varepsilon'|\varepsilon,a}$, grid $\{k_j\}$
Initialize $G_{t+1}(k_j, \varepsilon') = 0$ for all (j, ε')
for each (k_i, ε_j) with $G_t(k_i, \varepsilon_j) > 0$ **do**
  Compute optimal savings: $k' = g(k_i, \varepsilon_j, m_1, a_t)$
  Find bracketing grid points: $k' \in [k_J, k_{J+1}]$
  Compute lottery weight: $\omega = 1 - \frac{k' - k_J}{k_{J+1} - k_J}$
  for each next-period shock ε' **do**
    $G_{t+1}(k_J, \varepsilon') \ += \ \omega \cdot \pi_{\varepsilon'|\varepsilon_j, a_t} \cdot G_t(k_i, \varepsilon_j)$
    $G_{t+1}(k_{J+1}, \varepsilon') \ += \ (1-\omega) \cdot \pi_{\varepsilon'|\varepsilon_j, a_t} \cdot G_t(k_i, \varepsilon_j)$
  end for
end for
Output: Updated histogram G_{t+1}

Notation note. The interpolation weight is denoted ω throughout this deck and the script. It is the probability mass sent to the lower bracketing grid point.

Implementation: Young's Update in NumPy

Four scatter-add lines (one per leaf of the cascading fork):

```
def young_step(G, kp, pi_eps, k_grid):
    """G[i,j]: mass at (k_grid[i], eps_j); kp[i,j]: policy k'."""
    G_next = np.zeros_like(G)
    dk     = k_grid[1] - k_grid[0]
    n_k    = len(k_grid)
    for j in range(G.shape[1]):
        for i in range(G.shape[0]):
            x     = np.clip(kp[i, j], k_grid[0], k_grid[-1])
            J     = min(int((x - k_grid[0]) // dk), n_k - 2)
            omega = (k_grid[J + 1] - x) / dk            # in [0, 1]
            for jp in range(G.shape[1]):
                w = pi_eps[j, jp] * G[i, j]
                G_next[J,     jp] += omega       * w
                G_next[J + 1, jp] += (1 - omega) * w
    return G_next
```

JAX vectorisation of the same operator (`distribution_step` via `jax.vmap` and `.at[].add()`) is in `KrusellSmith_Tutorial_CPU.ipynb`.

Why Closed-Form Bracketing Matters on GPUs

The lookup. Young's update — and any piecewise-linear interpolation — needs the bracketing index for each query k':

$$J(k') \;=\; \max\{\, n : k_n \le k' \,\}, \qquad k' \in [k_J,\, k_{J+1}).$$

Generic approach. Binary search (e.g. `jnp.searchsorted`): $\mathcal{O}(\log N_k)$ per query, with data-dependent branches.

Consequences on GPU / under XLA.

- Warp divergence across SIMT threads taking different branches.
- Irregular gather patterns and extra kernel launches.
- Opaque search op breaks fusion with the surrounding interpolation arithmetic.

Closed-form alternative (uniform grid).

$$J(k') \;=\; \mathrm{clip}\left(\left\lfloor \frac{k'-k_0}{\Delta k} \right\rfloor,\; 0,\; N_k - 2\right).$$

One fused multiply–add and a cast: branch-free, uniform across threads, fuses with the lottery weight $\omega = (k_{J+1} - k')/\Delta k$ into a single GPU kernel. This is exactly the line `J = int((x - k_grid[0]) // dk)` on the previous slide.

Log-Spaced Grids Without Giving Up the Closed Form

Why log-spacing. The consumption policy is steepest near the borrowing constraint $k \to 0$; we want fine resolution there without wasting points in the right tail.

Recipe. Equidistant knots in a transformed coordinate $x = \log(c + k)$ for a small shift $c > 0$:

$$x_n \;=\; x_0 + n\,\Delta x, \qquad k_n \;=\; e^{x_n} - c, \qquad n = 0, \ldots, N_k.$$

Bracketing stays closed-form.

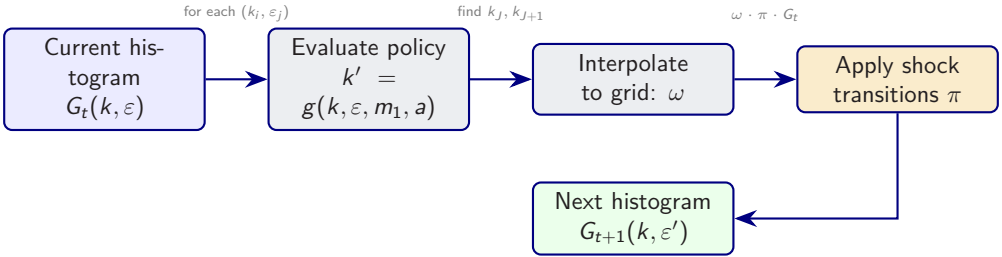
$$J(k') \;=\; \mathrm{clip}\left(\left\lfloor \frac{\log(c+k') - x_0}{\Delta x} \right\rfloor,\; 0,\; N_k - 2\right).$$

Still $\mathcal{O}(1)$ per query, no search, no branches.

Keep interpolation in k-space. The *index* is computed in x-space, but the lottery *weights* use the original k_n's, so the off-grid policy is piecewise linear in k — consistent with the on-grid consumption values and with Young's mean-preserving lottery.

Concrete implementation: `interpolate()` and `distribution_step()` in the Krusell–Smith JAX tutorial of Azinovic-Yang & Žemlička (2025), *Deep Learning in the Sequence Space*, arXiv:2509.13623.

Flow Diagram: One Forward Step



Key properties of each step:

- **Policy evaluation**: uses the solution from value function iteration (or neural net)
- **Interpolation**: linear weights ensure mean preservation
- **Shock transition**: probability-weighted mass redistribution across ε' states
- **Result**: G_{t+1} sums to 1 and has the policy-implied mean. At the fixed point this matches $\hat{H}(m_1, a)$; away from it, the outer loop closes the gap.

The Full Algorithm: Iterating to Equilibrium

Krusell–Smith Algorithm with Young's Simulation

Step 1: Initialize forecasting coefficients $A(a), B(a)$
repeat
  Step 2: Solve household problem via VFI given forecasting rule $\hat{H}$
  Step 3: Forward-simulate distribution for T periods via Young's method:
  for $t = 1, \ldots, T$ **do**
    Draw aggregate shock a_t from Markov chain
    Update histogram: $G_t \xrightarrow{\text{Young}} G_{t+1}$
    Record realized mean: $m_1^t = \sum_{i,j} G_t(k_i, \varepsilon_j) \cdot k_i$
  end for
  Step 4: Re-estimate $A(a), B(a)$ by OLS on simulated (m_1^t, m_1^{t+1}, a_t)
  Step 5: Check convergence: $|A^{\text{new}} - A^{\text{old}}| < \text{tol}$
until converged

PAUSE Finger Exercise: Histogram Mass Redistribution

Exercise (5 min)

Consider a 4-bin histogram on the grid $\{0.5, 1.0, 1.5, 2.0\}$ with initial masses $\mu = (0.25, 0.25, 0.25, 0.25)$. Suppose the policy function maps:

$$g(0.5) = 0.8, \quad g(1.0) = 1.3, \quad g(1.5) = 1.7, \quad g(2.0) = 1.9.$$

1. For $g(0.5) = 0.8$: which two grid points does 0.8 fall between? What are the interpolation weights?
2. Compute the updated mass μ' at grid point 1.0 after redistributing the mass from $g(0.5) = 0.8$ only (ignoring other bins for now).

Solution: Histogram Mass Redistribution

Answer

1. $g(0.5) = 0.8$ falls between grid points 0.5 and 1.0.
   Weight on 1.0: $\omega = \frac{0.8-0.5}{1.0-0.5} = 0.6$. Weight on 0.5: $1 - \omega = 0.4$.
2. The mass $\mu(0.5) = 0.25$ is split: grid point 1.0 receives $0.6 \times 0.25 = 0.15$, and grid point 0.5 receives $0.4 \times 0.25 = 0.10$.
   To get the full updated $\mu'(1.0)$, you would also add contributions from any other bins whose policy maps into the $[0.5, 1.5]$ interval—but this exercise only asked for the contribution from $g(0.5)$.

Total mass is preserved: $0.10 + 0.15 = 0.25 =$ original mass at bin 0.5. ✓

Convergence and Accuracy

Quality of the forecasting rule:

- $R^2 > 0.999$ for the log-linear rule $\hat{H}(m_1, a) = \exp(A(a) + B(a) \log m_1)$
- KS (1998) also tested 2- and 3-moment rules (adding variance, skewness): R^2 barely changes
- The one-moment benchmark is an empirical finding, not a modeling restriction

Accuracy tests (Den Haan, 2009):

- Compare simulated m_1 path with the forecasting rule prediction
- Maximum one-step-ahead forecast error $< 0.1\%$
- Simulation length: $T = 10{,}000$ periods (discarding first 500 for burn-in)

Convergence speed:

- Young's non-stochastic simulation removes MC noise from the simulated means
- Each iteration still includes VFI + forward simulation + OLS
- Outer-loop counts and runtime are tolerance- and implementation-dependent

Implementation: Grid Design

Practical considerations from Young (2010):

Value function grid:

- $\sim$150 points, irregularly spaced
- Finer near the borrowing constraint (high curvature)
- Coarser for large k (smooth policy)
- Used for VFI with cubic splines

Simulation grid:

- $\sim$5000 points, uniformly spaced
- Uniform spacing essential for smooth histograms
- Avoids spikes in the distribution
- Upper bound: $k_{\max} = 1200$ (large enough)

Important

Two separate grids: one for solving (irregular, fewer points), one for simulating (uniform, many points). The simulation grid must be fine and uniform to produce smooth distributions.

Boundary Cases

What happens at the grid boundaries?

- $k' \leq k_{\text{min}}$: all mass goes to the first grid point ($p = 1$)
- $k' \geq k_{\text{max}}$: all mass goes to the last grid point ($p = 0$)

Importance of grid bounds:

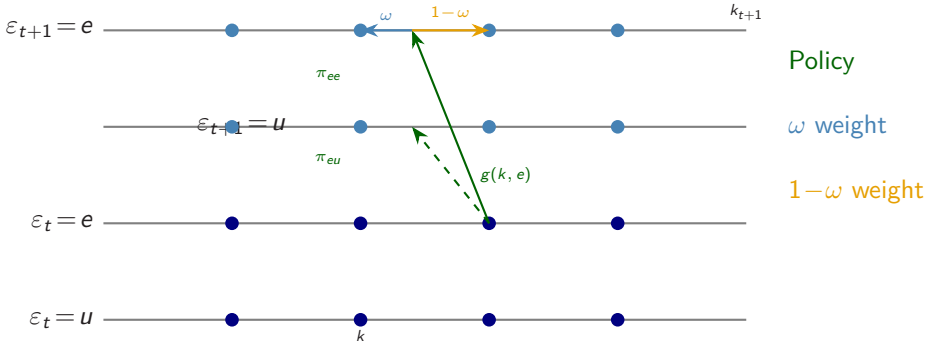
- If k_{max} is too small, mass "piles up" at the boundary $\Rightarrow$ mean is not preserved
- If k_{max} is too large, the grid wastes resolution in empty regions
- Rule of thumb: run a preliminary simulation, check that no mass reaches boundaries

From the code (Azinovic et al. 2022)

```
anext = tf.minimum(anext, amax * tf.ones_like(anext))
```

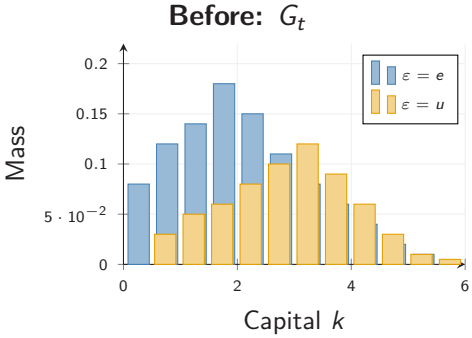
Asset holdings are clipped at $a_{\text{max}} = 20$ to prevent overflow.

Young's Simulation Procedure (Schematic)

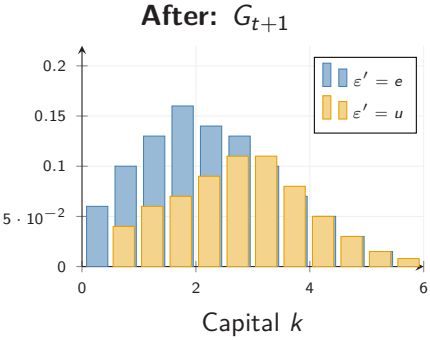


Arrows show: (1) policy maps current (k, ε) to k', (2) k' is split between grid points with weight ω, (3) mass transitions across ε' states with probabilities $\pi_{\varepsilon'|\varepsilon}$.

One Forward Step: Before and After



Mass concentrated at low k; borrowing-constrained agents at left edge



After policy + interpolation + shock transitions: mass slightly rightward

Mean preserved: $\bar{k}_t = \bar{k}_{t+1}$

Exact mean preservation!

Summary of Young's Method

Key Takeaways

1. Represents the distribution as a histogram on a fixed grid
2. Updates deterministically using policy functions + transition probabilities
3. Preserves the mean exactly; approximates higher moments
4. No sampling noise—faster and more accurate than Monte Carlo
5. Compatible with any solution method for the household problem (VFI, neural nets, . . .)

"Rather than use Monte Carlo simulation to generate an updated cross-sectional distribution . . . I use a histogram over a fixed grid to construct tomorrow's distribution."

— Young (2010, p. 38)

Part III

Benchmark: Krusell–Smith Logic

Benchmark: Krusell–Smith Logic

The Krusell–Smith (1998) economy:

- Continuum of ex ante identical, infinitely-lived agents
- Incomplete markets: no insurance against idiosyncratic risk
- Aggregate uncertainty: productivity shocks affect all agents
- Single asset: one-period bonds/capital with borrowing constraint

The challenge

Each agent must forecast future prices to make optimal savings decisions. But prices depend on the entire wealth distribution—which is an infinite-dimensional object.

$\Rightarrow$ Agents use a simple forecasting rule based on the mean of the distribution.

Model Switch: Benchmark vs. Teaching Implementation

Benchmark layer

- **Story model:** Krusell–Smith (1998)
- **Role here:** motivate approximate aggregation and the outer forecast loop
- **Young step:** deterministic distribution propagation inside KS

Teaching-code layer

- **Notebook 11:** Appendix A.5 endowment economy
- **Role here:** show Young's histogram as a DEQN state and simulator
- **Key change:** a price net replaces KS forecast-rule regression

Same operator, different shell

KS explains why distribution tracking matters. Appendix A.5 shows the same histogram update inside a DEQN.

Teaching Model: Preferences

Epstein–Zin recursive utility (separates risk aversion from IES):

$$\boxed{V(x_t) = \max_{b_{t+1} \geq 0} \left[(1-\beta)\, c_t^{1-\rho} + \beta\, \chi_t(V(x_{t+1}))^{1-\rho} \right]^{\frac{1}{1-\rho}}}$$

Certainty equivalent:

$$\chi_t(V_{t+1}) = \mathbb{E}\left[V_{t+1}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$

Parameter	**Symbol**	**Value**
Discount factor	β	0.95
Inverse of IES	ρ	2 (IES = 0.5)
Risk aversion	σ	8

Note: $\rho \neq \sigma \Rightarrow$ agents' intertemporal substitution and risk preferences are separated.

Teaching Model: Budget and Shocks

Endowment economy (as in Azinovic et al. 2022, Appendix A.5):

- Aggregate output: $Y_t = w(a_t)$ depends on aggregate shock a_t
- Idiosyncratic labor endowment: $\eta_t \in \{0.8, 1.2\}$ with transition matrix Π_η
- Aggregate shocks: 6 states (2 uncertainty $\times$ 3 income levels)

Budget constraint:

$$\boxed{c_t + p_t \cdot b_{t+1} = b_t + \eta_t \cdot w(a_t)}$$

- b_t: bond holdings (asset)
- p_t: bond price (endogenous, clears the market)
- $\eta_t \cdot w(a_t)$: individual labor income
- Borrowing constraint: $b_{t+1} \geq 0$

Teaching Model: Individual Problem

Each agent solves:

$$V(b, \eta, x^{agg}) = \max_{b' \geq 0} \left[(1-\beta)\, c^{1-\rho} + \beta\, \chi(V(b', \eta', x'^{agg}))^{1-\rho} \right]^{\frac{1}{1-\rho}}$$

subject to:

$$c = b + \eta \cdot w(a) - p \cdot b'$$

State variables:

Individual:

- b_t: current bond holdings
- $\eta_t \in \{0.8, 1.2\}$: idiosyncratic shock

Aggregate:

- $a_t \in \{1, \ldots, 6\}$: aggregate shock
- μ_t: wealth distribution (histogram)

The bond price p_t clears the market: $\int b_{t+1}\, d\mu_t = \bar{B}$; the teaching notebook normalizes $\bar{B} = 1$ (unit net supply).

Teaching Model: Equilibrium Objects

Recursive Competitive Equilibrium

1. **Agents optimize:** given bond prices, each agent chooses b' to maximize utility
2. **Market clearing:**

$$\int b_{t+1}(b, \eta, x^{agg})\, d\mu_t(b, \eta) = \bar{B}, \qquad \bar{B} = 1 \text{ in the notebook}$$

3. **Distribution evolves consistently:** $\mu_{t+1} = T(\mu_t, a_t)$ given optimal policies
4. **Rational expectations:** agents' forecast of future prices is self-confirming

The bond price p_t:

- Depends on the aggregate state (a_t, μ_t)
- Must be found such that aggregate demand = aggregate supply = 1
- In the DEQN approach: a neural network outputs p_t directly

Traditional KS Solution Method

Step-by-step algorithm:

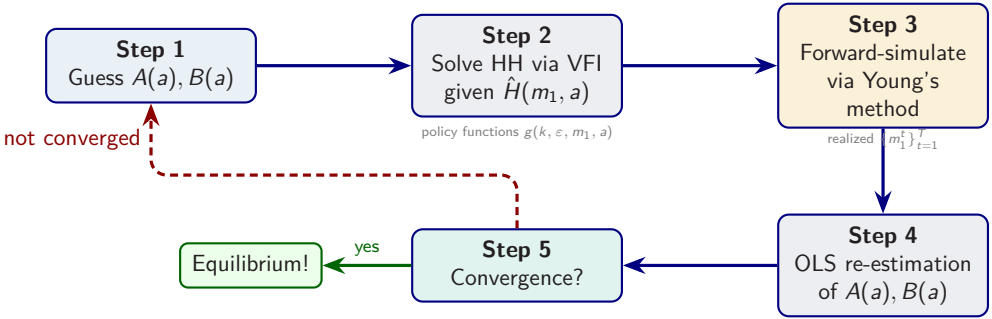
1. **Guess** forecasting rule: $p(a) \approx f(m_1, a)$
2. **Solve** household problem via VFI on (b, ε, m_1, a) grid
3. **Forward-simulate** distribution via Young's method for T periods
4. **Compute** realized m_1 and prices from simulated distributions
5. **Re-estimate** forecasting rule by regression on simulated data
6. **Check convergence**: if not converged, go to step 2

Limitations:

- VFI on fixed grid: limited resolution, curse of dimensionality
- Forecasting rule: restricted to low-order polynomial form
- Adding more moments/states is expensive

$\Rightarrow$ Can we do better with deep learning?

The Krusell–Smith Outer Loop



Young's simulation makes the forward-distribution step deterministic and fast, but total runtime still depends on the VFI tolerance, forecasting-rule tolerance, and implementation.